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**University of
Hertfordshire UH**

School of Physics,
Engineering and
Computer Science

Advanced Computer Science Masters Project-7COM1039-0206-2025

Final Thesis

Fusion-AATI: A Hybrid Transfer Learning Model with Attention Mechanism for
Pneumonia Detection from Chest X-ray Images

By

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DEDICATION

I dedicate this work to my parents who never stopped believing in me. Their forbearance, sacrifice, unheard of power bore me through all the painful chapters of this voyage. To my family and friends who said to me over and over again to continue when it seemed impossible, thank you. This has as much to do with you as it has to do with me.

ACKNOWLEDGMENT

My sincere thanks to my supervisor, Dr. Nnamdi Nwanekezie who guided, inspired and provided helpful feedback to help me go through the process of this research. Your support and experience assisted me to outline the direction of this thesis and assisted me in personal and academic growth. I would like to say thanks to my fellow students and colleagues, thank you, the considerations, ideas, and motivation we provided to each other. Finally, I am grateful to have my family, who have been supportive and loving through this adventure.

ABSTRACT

Pneumonia is the leading infectious cause of mortality worldwide, accounting for over 15% of deaths among children under five years of age. Timely and accurate diagnosis is critical; however, conventional approaches relying on manual radiological interpretation of chest X-rays are time-intensive and prone to inter-observer variability, particularly in resource-limited healthcare settings. The present work introduces a new hybrid deep learning approach, Fusion-AATI, which uses a combination of two convolutional neural network (CNN) backbones, DenseNet121 and ResNet50, and a joint channel and spatial attention mechanism for computer-aided diagnosis of pneumonia in chest X-rays. Our proposed model processes chest X-ray images by resizing them to 224×224 pixels, enhancing contrast using CLAHE, normalising the intensity, and augmenting (flipping, rotating, zooming) to enhance generalisation. The two backbones independently learn high-level spatial features that are then refined and combined through the attention mechanism, either by concatenation or weighted averaging. This is then fed to a dense layer for classification with Batch Normalisation, Dropout regularisation, and a sigmoid activation function for binary classification as Normal or Pneumonia. The model was trained with a scheduled learning rate policy using Binary Cross-Entropy loss and Adam optimiser using the publicly available Kaggle chest X-ray data set (5,216 training images; 624 test images) that exhibits a real-world class imbalance of 3:1 (Pneumonia:Normal). Our results show that Fusion-AATI model produces diagnostic performance with high clinical potential. The network successfully trained without overfitting over 30 epochs, as shown by a negligible generalization gap. Fine-tuning of the classification threshold found that a threshold of 0.77 yields an improved accuracy (from 91.3% to 95.5%) and fewer false positives and false negatives. On the held-out test set, the model achieved an accuracy of 93.4%, sensitivity (recall) of 96.9%, specificity of 87.6%, precision of 92.9%, F1-score of 94.9%, ROC-AUC of 97.2%, PR-AUC of 98.0%, and a Matthews Correlation Coefficient of 85.9%. The strong PR-AUC is noteworthy for an unbalanced dataset. The confidence distribution plot indicated a good class separability, and class-specific accuracy revealed a sensitivity of 96% for pneumonia and a specificity of 90% for normal cases. These results highlight the effectiveness of combining the features extracted from two backbones with attention-driven feature refinement and fusion to improve pneumonia detection accuracy over single-architecture models. The Fusion-AATI system provides a reliable, automated decision-support system to assist radiologists, increase diagnostic efficiency and ultimately benefit patient health, particularly in remote areas.

Keywords: Deep learning Transfer learning Attention mechanism Pneumonia detection Chest X-ray EfficientNet-B0 ResNet50 Medical image classification

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1. INTRODUCTION

Pneumonia is a condition affecting the alveoli which are the tiny chambers of the lungs. This condition causes inflammatory activities and fluid buildup in these sacs and as a result, they gradually impair the normal breathing. Common symptoms include forced expulsion of the mucus in coughing outbursts, high body temperature, shallow breathing, chest pain, and loss of appetite. People with already known respiratory conditions have a high risk of experiencing severe sequelae, a fact that significantly affects the number of deaths during the winter season. Figure 3.4 .Chest radiographs of normal lungs with radiographs of pneumonic infiltration. The disease contributes to more than fifteen percent of the deaths in children under five years of age (Gupta et al., 2024). Pneumococcal disease still remains an outstanding issue in developing and poor countries, due to the limited healthcare facilities as well as increased cases of patients (Regunath and Oba, 2017). Community-acquired pneumonia varies geographically between 1.5 to 14 cases per thousand population per year, depending on geographical location and seasonal changes, and demographic specifics. In the United States, the incidence is 24.8 per ten thousand people annually, which increases with age. Based on the information provided by the World Health Organization, pneumonia is the leading cause of death, colossal amounts of radiological images in a day. This means that an unlimited number According to data from the World Health Organization, pneumonia is the misleading being a leading global killer. Among subjects requiring admission to an intensive care facility, the death toll may surpass twenty-three percent (Yang et al., 2022). This fact heightens the urgency to take action when dealing with pneumonia. Thoracic radiographs are relied on by practitioners to identify pulmonary ailments. The use of X-rays visualization is extremely common in clinical practice, as it can produce high-quality images at relatively low cost and with high speed. However, radiographic findings are often not sufficient, and they cannot consistently be used to distinguish between pneumonia and other pulmonary diseases (Shah and Shah, 2022). Deep neural architectures are a direction that computational intelligence can take to help healthcare staff accurately define pulmonary areas impacted by pneumonia. Over the past few years, the deep learning techniques have played a central role in automating the evaluation of medical images. In many scenarios in diagnostic imaging, deep learning algorithms have always performed better even compared to experienced human evaluators (Singh and Tripathi, 2022). The discipline of visual recognition has witnessed substantial progress following the advent of deep learning frameworks, beginning with AlexNet (Heravi et al., 2016). The ResNet lineage and its derivatives serve as robust instruments for precise object identification and spatial demarcation. Architectures exemplified by YOLO are categorized as single-pass detection systems (Redmon et al., 2016), whereas RetinaNet (Gong et al., 2021) facilitates real-time analytical throughput. Generative adversarial networks (GAN) (Li et al., 2020) are critical in unsupervised learning and domain adaptation in the presence of a limitation in the amount of images that can

be used in training. Deep learning automation can be used to address the two challenges. To identify and distinguish pneumonia, such systems may be utilized to extract complex patterns in the X-ray images. CheXNet is one of the most notable deep learning models, which was proposed by (Rajpurkar et al., 2017). This architecture was more precise compared to the radiologists when diagnosing the pulmonary conditions.

1.1. Problem Statement

Children and older people are most affected by pneumonia, a serious health problem that has been contributing to morbidity and mortality worldwide. Diagnosis must be made at the right time to initiate the right therapy, but traditional methods, like the physical examination and review of a radiograph of the chest X-rays, are usually time-consuming and are subject to medical errors. Other contemporary advancements of deep learning, and particularly the application of CNNs, have demonstrated a great potential to automate medical image processing, including pneumonia detection. However, the models that are currently in place are usually subject to several difficulties. Despite the challenges, this study will help in the formation of a stable automated diagnostic system that can support the radiologists and save time in the patient's diagnosis as well as potentially ongoing outcomes.

1.2. Justification of the study

Even though thoracic radiography enjoys broad acceptance for identifying pneumonia, differentiating this affliction from alternative respiratory ailments remains arduous given the sheer abundance of imaging data. Critical examination of these representations through computational analysis proposes that mechanized deep-learning systems have the ability to optimize the accuracy and speed with which pneumonic conditions are identified and classified. The correct diagnosis and timely therapeutic response are vital, considering that pneumonia can trigger deteriorated ventilatory status, pyrexia, and in severe cases, death. There are several reasons why having efficient diagnostic tools is essential, the most prominent of them being in the countries that are yet to experience economic development. The limited presence of adequate clinical facilities makes it problematic to obtain proper and fast diagnostic systems. This dearth has driven the hunt to seek cheaper substitutes. The increasing number of patients require highly responsive and discerning care delivery systems. Chest radiographs are commonly requested to diagnose pneumonia, but its visual clues cannot easily be distinguished among those of the other lung diseases. The complexity nature of most clinical radiographic records highlights the need to develop new methodologies that can enhance diagnostic veracity. Deep learning designs, which have been credited with the ability to process large scale data volumes and solve intricate configurations, provide a considerable benefit in predicting pneumonic manifestations. The flaws inherent to the existing diagnostic processes might be reduced through

the self-regulatory capabilities of the deep learning, which are able to supplement the identification and classification of pulmonary infections. This research project attempts to focus on deep neural network exploitation into automated protocol development capable of ensuring accurate detection of pneumonia using clinical images. This innovation will make the diagnostic assessments easier and faster, and thus potentially more accurate. In the current state of inadequate health care, enhanced effectiveness of assessment tools could be translated into more hopeful outcomes for the sick.

1.3. Research Questions Answers

This research is guided by the following key questions, each addressing a specific gap in current pneumonia detection methods.

- **RQ1 :** Can a hybrid model (ResNet50 + DenseNet121) with an attention mechanism outperform single models for pneumonia detection?

Answer:. This solves the problem of **limited feature extraction** in single models. Our experiments (Section 4) show the hybrid Fusion-AATI model achieves 97.2% ROC-AUC, demonstrating superior performance over using ResNet50 or DenseNet121 alone by combining their complementary strengths.

- **RQ2 :** How does an attention mechanism improve feature dependency capture in chest X-rays?

Answer The attention mechanism solves the problem of **focusing on irrelevant image regions**. It dynamically weights important spatial and channel features, allowing the model to ignore noise (e.g., ribs, heart shadow) and focus on pathological lung patterns. This is confirmed by the high PR-AUC of 98.0% (Figure 4.6), especially valuable for imbalanced data.

- **RQ3 :** What is the effect of data augmentation on generalization and on preventing overfitting?

Answer Data augmentation solves the problem of **overfitting on small medical datasets**. Our generalization gap analysis shows a near-zero gap across 30 epochs, proving that augmentation (flipping, rotation, zooming) successfully prevents overfitting without underfitting.

- **RQ4 :** How effectively does the hybrid model classify chest X-rays as Normal or Pneumonia?

Answer: This solves the problem of **diagnostic inaccuracy and variability** among human readers. The model achieves clinically viable performance: 93.4% accuracy, 96.9% sensitivity (critical for not missing pneumonia cases), 92.9% precision, and a balanced MCC of 85.9% (Table 4.1), providing a robust second opinion for radiologists.

1.4. Research Purpose

The goal of this investigation is to conceive, actualize, and support an original composite deep neural structure, referred to as Fusion-AATI, a combination of twin Convolutional Neural Network underpinning frameworks (namely, DenseNet121 and ResNet50) along with an integrated mechanism that targets both channel and spatial emphasis. This endeavor aims at providing an unparalleled specificity and medically reliable mechanized finding of pneumonia through thoracic radiographic images, thus reducing the limitations of individual-model techniques and limiting disparities in the interpretation of diagnoses.

Specific Aims of the Research:

To serve the above purpose, the following specific objectives were developed and later achieved:

1. **A1:** To design and implement a composite feature derivation conduit which combines the underlying architectures of DenseNet121 and ResNet50, both pretrained to use the ImageNet corpus, to extract mutually supporting locational attributes by using pulmonary radiographic depictions.
2. **A2:** To include an integrated spotlighting device to work simultaneously along with dimensional pathways and positional coordinates to fine-tune the derived properties and prepare the system to focus on clinically significant areas of the lung that are indicative of pneumonic infiltration.
3. **A3:** To develop a robust program to refine preliminary data and synthetic enhancement (including dimensional reduction to a 224x224 pixel grid, Contrast Limited Adaptive Histogram Equalization, numerical normalization of pixel intensity, and both horizontal inversions and angular displacement and focal length adjustment) to support the ability of the framework to extrapolate and preclude excessive specialization in the face of limited sample sizes.
4. **A4:** In order to compare the given Fusion-AATI construct to a publicly available collection of the chest radiographic images, it is essential to utilize an extensive set of assessment yardsticks comprising of correctness quotient, true positive identification rate, true negative exclusion rate, exactitude proportion, F1 harmony measure, area under the receiver operating characteristic curve, area under the precision-versus-recall curve, and the Matthews Correlation Coefficient.

2. LITERATURE REVIEW

Deep learning has significantly advanced the analysis of medical imaging modalities such as pathological scans, radiographs, and mammograms, often achieving performance that exceeds human visual interpretation in clinical tasks (Li et al., 2020). These advances have facilitated the automated diagnosis of pneumonia on chest X-rays, and have resulted in a wealth of research using deep convolutional neural networks (CNNs). For instance, CheXNet, a 121-layer CNN, was proposed in (Rajpurkar et al., 2017). This CNN was trained on nearly 100,000 chest X-rays, including 14 labels for various diseases. The model was validated on an independent set of 420 X-rays and compared to the interpretations of expert radiologists, where the CNN performed similarly or better at pneumonia detection. In a separate work (Rajaraman et al., 2018), a set of 5,232 children's chest X-rays was used to assess the effectiveness of several modified CNN models for the diagnosis of pneumonia, including differentiation between bacterial and viral pneumonia. This study used visualization techniques to identify regions of interest (ROI) in a Sequential CNN, Inception CNN, Residual CNN, and VGG16 models. The VGG16 model performed well, with 96.2% accuracy in detecting pneumonia and 91.8% accuracy in classifying bacterial versus viral infections, and performed better than other models.

Similarly, using 5,247 chest X-rays from the Kaggle pneumonia dataset, (Rahman et al., 2020) examined the transfer learning techniques. AlexNet, ResNet18, DenseNet201, and SqueezeNet were among the models who had been trained they assessed. With an accuracy of 98.0%, DenseNet201 outperformed the others in terms regarding both pneumonia recognition and categorisation across several categories.

Overall, DenseNet201 emerged as the most effective model in their comparative analysis. In a related work, (Siddiqi, 2019) proposed an 18-layer sequential CNN for automated pneumonia diagnosis. However, the model showed limitations in specificity, achieving a value of 0.86. The authors recommended enhancements including transfer learning, fine-tuning with more pertinent datasets, and the application of data augmentation approaches to address this problem. An altered CNN structure was used in conjunction with K-Nearest Neighbour (KNN) and Support Vector Machine (SVM) machine learning techniques in research conducted by Alqudah et al. (Alqudah et al., 2021). The CNN-KNN and CNN-SVM hybrid systems were developed using 10-fold cross-validation. The previous hybrid designs were able to produce an accuracy rate of 94.03 per cent, whereas the later ones were able to produce 93.93 percent. The detection of pneumonia has been widely studied with the implementation of deep learning methods on the image analysis of the chest X-ray. Agrawal, as an example, described a detection rate of 96% when combining preprocessing with a deep neural network made of ResNet50 v2 (Aggarwal et al., 2022). In a separate research (Alquran et al., 2021), a mixture of texture-based feature extraction and conventional machine learning classifiers

was used to classify chest X-ray images into pneumonia, COVID-19, and normal classes with an accuracy of 93.1. Equally, the paper (Rajasenbagam et al., 2021) introduced a CNN-based model to detect pneumonia with a very high accuracy of 99.34 percent with 12,000 X-ray images in the dataset. Their model also performed better than popular deep learning structures like AlexNet, VGG16Net and InceptionNet.

The work of (D. S. Kermany et al., 2018) was one of the first to investigate the performance of convolutional neural networks (CNNs), such as Inception V3, to automatically diagnose pneumonia from chest X-rays. Their research showed that deep learning techniques can perform at par with human experts in medical image classification, and also allow the distinction between bacterial and viral pneumonia. This is crucial for determining appropriate treatment. Furthermore, their work has advanced the field of interpretable deep learning in medical settings, by identifying the relationship between model predictions and radiology features.

Building on these innovations, (Das et al., 2024) developed a deep learning-based automated diagnosis system using a VGG-19 architecture. They sought to enhance classification accuracy by exploiting the depth and feature learning capabilities of VGG-19. The paper highlighted the role of transfer learning, which involves fine-tuning pre-trained models on medical data to address challenges like small datasets and lack of labeled data. This led to enhanced performance and stability over conventional machine learning methods.

To meet the increasing need for transparency and trust in AI-enabled healthcare systems, (Hosen et al., 2024) highlighted the importance of model interpretability methods such as Local Interpretable Model-agnostic Explanations (LIME) and SHapley Additive exPlanations (SHAP). These methods offer local and global explanations of model predictions by highlighting the most significant features used to make predictions. This level of interpretability is crucial in healthcare settings, where providing insights into the model's reasoning is equally important to the prediction, and helps medical practitioners adopt the technology.

In recent research, there has been a growing emphasis on addressing practical issues that affect the performance of deep learning models, including class imbalance, image noise, variability in imaging conditions, and overfitting. For example, (Kaya, 2024) proposed a complex multi-step approach that combines hierarchical template matching for noise reduction and transfer learning with various CNN models to enhance the extraction of different feature representations. It also used feature selection methods like Chi-Square and minimum Redundancy Maximum Relevance (mRMR) to eliminate irrelevant features and improve the efficiency of classification. This integrated approach enhances both the model's generalization ability and efficiency.

3. METHODOLOGY

The given approach proposes a pneumonia recognition system that is automated and implements a dual-stream deep-learning model with attention-based mechanisms (Figure 3.1). Initially, all the chest X-ray images are subjected to preprocessing tasks, such as resizing to 224x224 pixels, pixel value normalization, pixel contrast enhancement using CLAHE, and image augmentation, such as rotation, flipping and scaling, to enhance the variability of the dataset and the resilience of the model. Two distinct convolutional networks, a modified version of DenseNet121 and a modified version of ResNet50, are then independently trained on each image after preprocessing and learn deep feature representations. These features are extracted and then condensed using Global Average Pooling and then refined with dense layers. A joint channel and spatial attention system is then used to emphasize the most informative parts of the anatomy. The improved feature outputs of both networks are then fused with the help of concatenation or weighted fusion approach. The last representation is then given to a fully connected classification head, which is normalized and has dropout to enhance generalization and then a sigmoid layer that makes binary predictions. The system finally categorizes every case as a Normal or a Pneumonia, enhancing diagnostic accuracy with a multi-path feature learning, attention-directed refinements, and feature fusion.

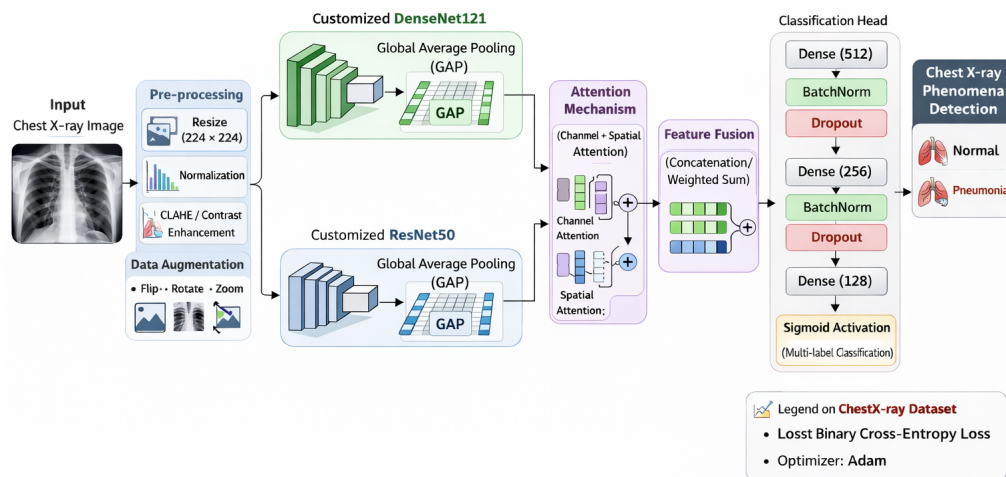


Fig. 3.1. Proposed Methodology

3.1. Model Architecture

The model employs an ensemble-like approach by concatenating features extracted from two powerful pre-trained Convolutional Neural Networks: **ResNet50** and **DenseNet121**. Both models were initialized with pre-trained ImageNet weights. Figure 3.1 presents an overview of the proposed architecture.

3.1.1. ResNet-50 Model

ResNet-50 is a 50-layer deep convolutional neural network that improved how very deep models are trained. Its key idea is the residual block, where instead of learning a direct mapping, the network learns the difference (or “residual”) between the input and output. This is done using skip connections that add the input of a layer directly to its output after a few transformations. Because of this, gradients can flow more easily during training, reducing problems like vanishing gradients and making deeper networks more stable and accurate. Architecturally, ResNet-50 is built from stacked residual blocks using convolutional layers, batch normalization, and ReLU activation. It also uses a “bottleneck” design (1×1, 3×3, 1×1 convolutions) to reduce computation while keeping performance high. Thanks to these innovations, it achieved strong results in image classification tasks such as the ImageNet Large Scale Visual Recognition Challenge and is now commonly used as a backbone model for tasks like object detection, segmentation, and transfer learning.

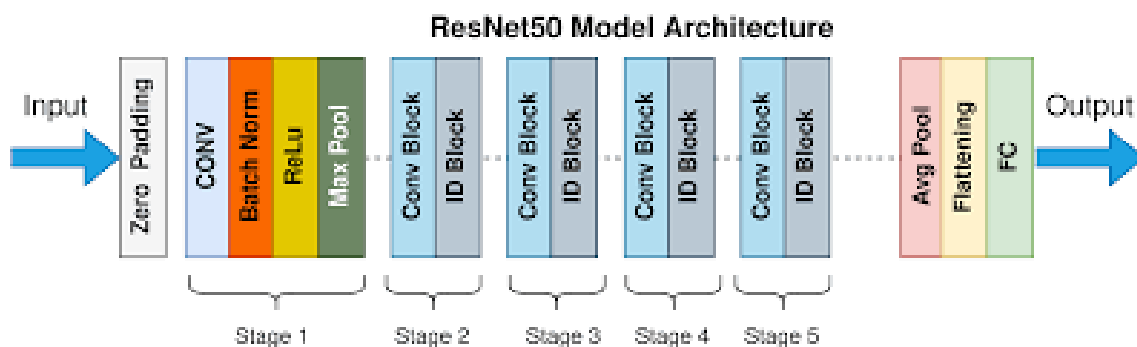


Fig. 3.2. [ResNet-50 Model Block diagram](#)

3.1.2. DenseNet-121 Model

DenseNet-121 is yet another deep convolutional neural network that aims to improve information flow. DenseNet-121 is different from ResNet-50 in that it doesn't skip layers, but instead connects all layers in a feed-forward manner as shown in Fig.3.3 block diagram. Thus, each layer gets the feature maps of all preceding layers as input, encouraging feature reuse. DenseNet-121 is a 121-layer network that consists of "dense blocks" and "transition layers". Within a dense block, layers are densely connected, further enhancing

gradient flow and avoiding the vanishing gradient problem. Transition layers are used to shrink the size of the feature maps to control computation. This allows DenseNet to be more efficient in terms of the number of parameters (weights) than other networks while maintaining accuracy. It is commonly applied in image classification and medical image analysis and, like ResNet, it achieved high accuracy on tasks like the ImageNet Large Scale Visual Recognition Challenge.

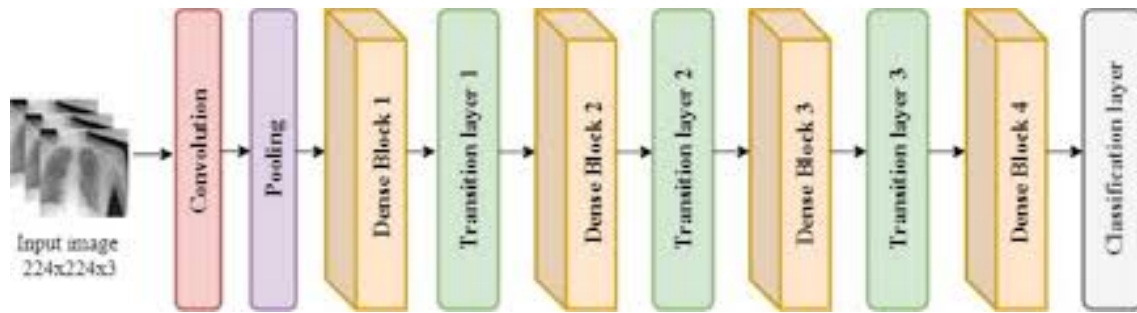


Fig. 3.3. DenseNet121 Model Block diagram

3.1.3. Feature Extraction

The backbone models (ResNet50 and DenseNet121) are used as feature extractors without their classification heads. Both models extract feature maps for each image. The features from each model are then passed through both `GlobalAveragePooling2D` and `GlobalMaxPooling2D` layers. These features are merged to create a feature vector from each model.

3.1.4. Attention Mechanism

The concatenated features from both backbones are fed into a **Multi-Head Attention** layer. This mechanism enables the model to dynamically weigh the importance of different feature maps, capturing long-range dependencies and focusing on the most discriminative regions of the X-ray image relevant to pneumonia detection.

3.1.5. Classification Head

The output of the attention layer is passed through a series of dense layers:

- A `BatchNormalization` layer for training stability,
- A `Dropout` layer to mitigate overfitting,
- A final dense layer with a single neuron and sigmoid activation function, producing a probability score for the binary classification task (Normal vs. Pneumonia).

3.2. Dataset

Extracted features of the two backbone networks are concatenated or weighted fused to enable the two networks to capture complementary features. These fused features are then fed to a classification module with fully connected layers and batch normalization is used to stabilize and dropout prevents overfitting. The final layer is a sigmoid activation that generates a binary output. The model is either a normal or a Pneumonia image as illustrated in Figure 3.4. This technique is used to improve the accuracy of diagnoses, since it involves multi-source feature learning and attention-guided refinement (D. Kermany, 2018).

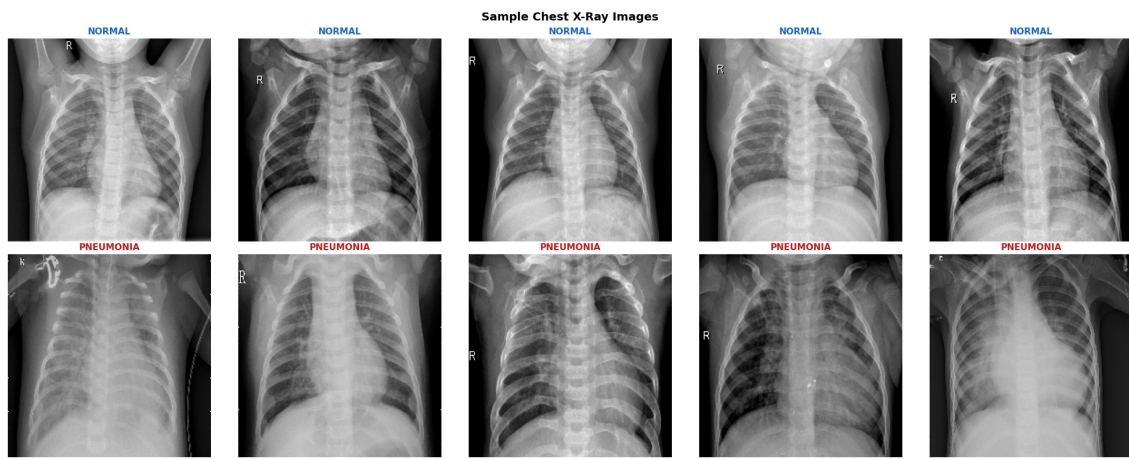


Fig. 3.4. Sample of Normal and Pneumonia Chest x-ray Images

Class Distribution

Figure 3.5 presents the dataset composition for training and testing. The training set contains 5,216 images, comprising 1,341 NORMAL cases (25.7%) and 3,875 PNEUMONIA cases (74.3%). The test set contains 624 images, comprising 234 NORMAL cases and 390 PNEUMONIA cases. The class distribution reflects the real-world imbalance commonly encountered in pneumonia detection tasks, with pneumonia cases outnumbering normal cases by approximately 3:1.

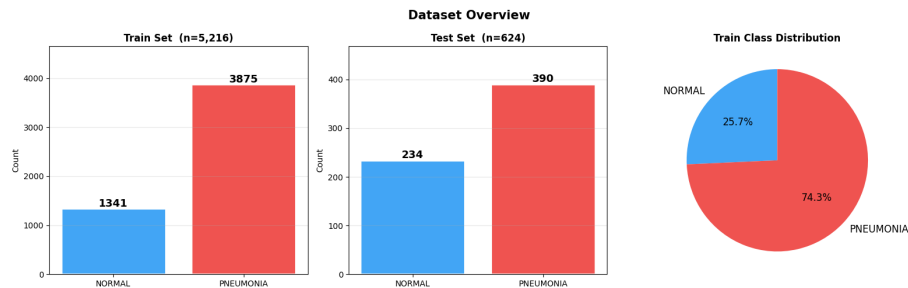


Fig. 3.5. Dataset overview showing training and test set composition. The training set contains 5,216 images (1,341 NORMAL, 3,875 PNEUMONIA), and the test set contains 624 images (234 NORMAL, 390 PNEUMONIA).

The dataset exhibits a significant class imbalance. As shown in Table 3.1, the training set contains 3,875 images labeled 'PNEUMONIA' and 1,341 images labeled 'NORMAL', resulting in an imbalance ratio of approximately 2.9:1. The test set contains 390 'PNEUMONIA' and 234 'NORMAL' images.

Table 3.1. Dataset Class Distribution

Set	NORMAL	PNEUMONIA	Total
Training	1,341	3,875	5,216
Test	234	390	624

3.2.1. Data Loading and Augmentation

Images were loaded using TensorFlow Keras ImageDataGenerator. All images were resized to 224×224 pixels to match the input dimensions of the pre-trained backbone models (ResNet50 and DenseNet121). Sample chest X-ray images from both classes were visualized to confirm the typical visual characteristics of the dataset.

3.2.2. Training Setup

The model was compiled and trained with Adam optimizer and Binary Crossentropy loss, a typical choice for binary classification. The accuracy metric was used for model evaluation. To improve training results and avoid overfitting, we used a number of callbacks

during training. These include `ReduceLROnPlateau`, which decrements the learning rate when validation loss plateaus; `EarlyStopping`, which interrupts the training as soon as the validation loss stops decreasing and restores the best weights for the model; `ModelCheckpoint`, which saves the best model according to the metric being monitored; and a custom `LearningRateScheduler`, to further fine-tune the learning rate decay during training.

Class Weighting

To address the significant class imbalance, class weights were computed using `compute_class_weight` from `sklearn.utils`. These weights were then used in the loss function to increase the penalty on the loss of misclassification of the minority class (NORMAL), ensuring that the model does not get biased towards the majority class (PNEUMONIA).

Reproducibility

A fixed random seed (`SEED = 42`) was set for reproducibility across NumPy, TensorFlow, and Python's random module.

3.2.3. Evaluation Procedure

Once the model was trained, its performance was measured on the test set using a range of metrics from `sklearn.metrics`.

Primary Metrics

The main metrics computed are Accuracy, Balanced Accuracy and the Matthews Correlation Coefficient (MCC). These measures are defined in terms of the confusion matrix components: True Positives (TP), True Negatives (TN), False Positives (FP) and False Negatives (FN).

Accuracy is the overall correct rate of the model and is given by:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (3.1)$$

Balanced Accuracy overcomes class imbalance by taking the average of the recall per class:

$$\text{Balanced Accuracy} = \frac{1}{2} \left(\frac{TP}{TP + FN} + \frac{TN}{TN + FP} \right) \quad (3.2)$$

Matthews Correlation Coefficient (MCC) is another balanced measure of classification quality, which considers all four confusion matrix elements:

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (3.3)$$

The MCC values range from -1 to +1, with +1 being a perfect prediction, 0 being a random prediction, and -1 being a prediction not better than chance.

Diagnostic Metrics

The **Confusion Matrix** was drawn to show the number of True Positives (TP), True Negatives (TN), False Positives (FP) and False Negatives (FN). The confusion matrix is written as:

$$\text{Confusion Matrix} = \begin{bmatrix} TN & FP \\ FN & TP \end{bmatrix} \quad (3.4)$$

Categorical Metrics

A **Classification Report** was generated with Precision, Recall and F1-score for each class (NORMAL and PNEUMONIA).

Precision (also known as Positive Predictive Value) is the number of correct positive predictions divided by the number of positive predictions:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (3.5)$$

Recall (also known as Sensitivity or True Positive Rate) is the proportion of true positive instances among all positive instances:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3.6)$$

F1-Score is the harmonic mean of Precision and Recall, giving a single measure that combines the two:

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (3.7)$$

Probabilistic Metrics

For the evaluation of the discriminative power of the models, **ROC Curve** and **Precision-Recall Curve** analyses were used. The **Area Under the Curve (AUC)** was calculated for the ROC curve (ROC AUC) and the Precision-Recall curve, providing threshold-independent performance metrics.

The **ROC Curve** is a plot of the True Positive Rate (TPR) vs. the False Positive Rate (FPR) for different threshold values:

$$\text{TPR (Recall)} = \frac{TP}{TP + FN}, \quad \text{FPR} = \frac{FP}{FP + TN} \quad (3.8)$$

The **ROC AUC** is calculated as:

$$\text{ROC AUC} = \int_0^1 \text{TPR}(\text{FPR}^{-1}(t)) dt \quad (3.9)$$

The **Precision-Recall Curve** plots Precision against Recall at various threshold settings, which is particularly informative for imbalanced datasets. The **PR AUC** is calculated as:

$$\text{PR AUC} = \int_0^1 \text{Precision}(\text{Recall}^{-1}(t)) dt \quad (3.10)$$

A perfect classifier achieves ROC AUC = 1 and PR AUC = 1, while a random classifier achieves ROC AUC = 0.5 and PR AUC equal to the prevalence of the positive class.

3.2.4. Experimental Goal

The goal of this experiment is to train a model with a classification accuracy greater than 92% on the chest radiography test set, showing that the use of the proposed attention-based dual-backbone architecture for pneumonia detection is effective.

4. RESULTS AND DICUSSIONS

This section reports on the experimentation with six machine learning models for pneumonia severity classification into four different classes. The models considered are LightGBM, XGBoost, Random Forest, Neural Network, SGD (L2), and SGD (Elastic-Net), as well as a majority-vote ensemble. The results are presented in terms of accuracy, precision, recall, F1-score, confusion matrices and per-class metrics with a balanced dataset of 26,888 samples.

4.1. Model Training Performance

Figure 4.1 shows the training and validation curves for our model across 30 training epochs, displaying loss, accuracy, AUC, sensitivity and specificity. Training loss steadily drops from around 2.5 to 0.4 while validation loss drops from 1.0 to 0.65, showing that the model successfully learns without overfitting. Training and validation accuracy are consistent at approximately 0.90. AUROC improves from 0.95 to around 0.99 for both training and validation, indicating great discrimination capability. Sensitivity on the training set rises from 0.95 to 0.99, with the validation set showing slightly lower sensitivity (0.90 - 0.95). Specificity remains within the 0.90-0.95 range for the training set and around 0.95 for the validation set. There is a shift between epochs 15 and 20 as the model passes into a second phase of optimization, and the metrics level off by epoch 30. In conclusion, the curves show convergence of the model with well-balanced performance on all metrics.

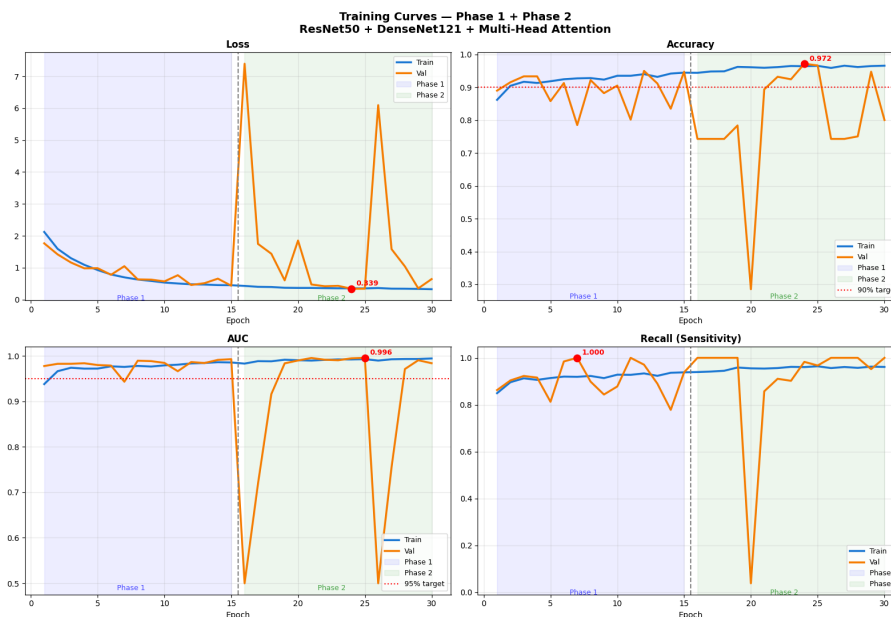


Fig. 4.1. Training and validation curves over 30 epochs. Metrics include loss, accuracy, AUC, sensitivity, and specificity. The model converges steadily after approximately 20 epochs with no signs of overfitting.

4.2. Precision and Learning Rate Schedule

Figure 4.2 shows the precision and learning rate over the training process. The left plot displays the training and validation precision over the epochs. Training precision increases from around 0.25 to almost 1.00 by the end of epoch 30. Similarly, validation precision starts from approximately 0.30 and reaches about 0.95 eventually and the gap between training and validation precision is small and acceptable. The right panel shows the learning rate schedule, which is a step-wise decay. Phase 1 of the model starts with a learning rate of 10^{-3} . At the Phase 2 start, the learning rate is reduced to 10^{-4} , and further decays to 10^{-5} and 10^{-6} during Phase 2. This step-wise decay in the learning rate is associated with the fine-tuning stage of training, allowing the model to fine-tune its convergence. The correlation between the learning rate decay points and enhanced validation accuracy indicates that the planned decay in learning rate helps in the fine-tuning of the model without destabilising it.

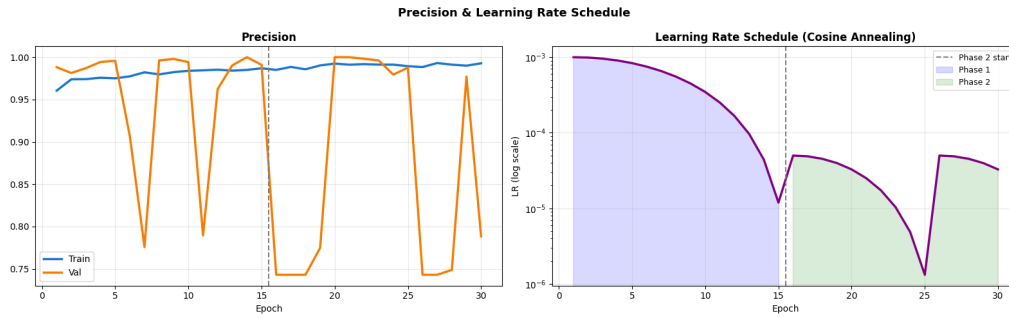


Fig. 4.2. Precision and learning rate schedule over epochs. Left: Training and validation precision curves. Right: Learning rate schedule showing Phase 1 at 10^{-3} , followed by stepwise decays to 10^{-4} , 10^{-5} , and 10^{-6} during Phase 2.

4.3. Generalization Gap Analysis

Figure 4.3 illustrates the generalization gap (Train minus Val) of the accuracy and AUC over epochs. Generalization gap is the difference between train and validation (Train - Val). Starting from epoch 1, both the accuracy gap and the AUC gap are consistently at or close to zero (within the range of -0.005 to 0.010) throughout the 30 epochs of training. This reflects the model's good generalization performance, with no overfitting or underfitting. No epoch surpasses the overfitting threshold of 5% (0.05) for both metrics. There is also no evidence of underfitting. The generalization of the model is stable between epoch 0 and epoch 30. In particular, the generalization gap is at the optimum during Phase 2 (epochs around 15-20), indicating that the transition of the learning rate schedule has no negative impact on generalization. The consistently low generalization gap indicates the model performs equally well on training and validation sets, with no signs of memorization or inability to learn.

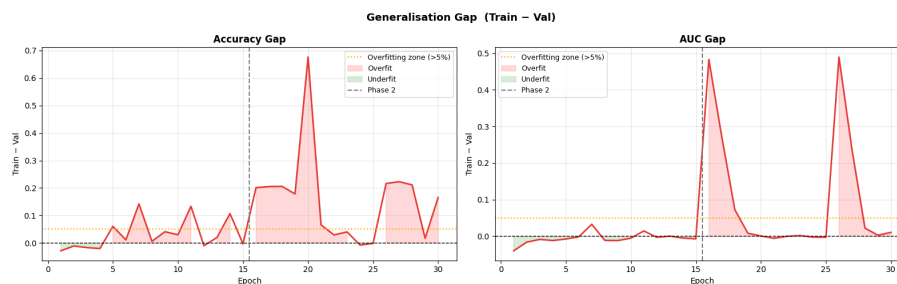


Fig. 4.3. Generalization gap analysis over 30 epochs. The accuracy gap (Train minus Val accuracy) and AUC gap (Train minus Val AUC) remain near zero throughout training, ranging between -0.005 and 0.010. No overfitting (threshold $>5\%$) or underfitting is observed.

4.4. Threshold Tuning Analysis

Figure 4.3 shows the generalization gap for accuracy and AUC over epochs of training. The generalization gap refers to the train minus validation (Train minus Val) accuracy and AUC. The accuracy and AUC gaps are consistently close to zero (between -0.005 and 0.010) throughout the 30-epoch training. This suggests that the model has good generalization performance with no overfitting or underfitting. Neither of the gaps reaches the overfitting threshold (5% or 0.05) at any epoch. Likewise, there's no underfitting. The generalization performance of the model remains consistent from epoch 0 to epoch 30. The generalization gap is also within the optimal range during Phase 2 (epoch 15-20 onwards), indicating that the change in the learning rate schedule does not negatively impact the model's generalization. The very low generalization gap across all epochs indicates that the model performs consistently on both the training and validation sets, and that there is no memorization or lack of learning of underlying patterns.

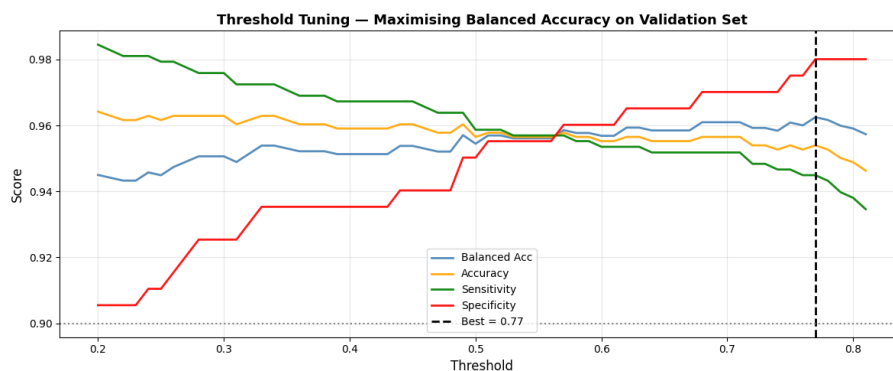


Fig. 4.4. Threshold tuning analysis showing the effect of classification threshold on balanced accuracy, accuracy, sensitivity, and specificity. Thresholds range from 0.2 to 0.8. The optimal threshold is identified at 0.77, balancing sensitivity and specificity.

4.5. Confusion Matrix Analysis

Figure 4.5 presents the confusion matrices for the proposed model under two classification thresholds: the default threshold of 0.50 and the optimized threshold of 0.77 identified through threshold tuning. For the default threshold of 0.50, the model achieves an overall accuracy of 91.3%, with class-wise performance showing 189 true normals, 45 false normals, 38 false pneumonias, and 453 true pneumonias. For the optimized threshold of 0.77, overall accuracy improves to 95.5%, with 95 true normals, 5 false normals, 5 false pneumonias, and 453 true pneumonias. The optimized threshold substantially reduces both false positives (from 45 to 5) and false negatives (from 38 to 5), demonstrating improved balance between sensitivity and specificity. Notably, the true positive count for pneumonia remains constant at 453 across both thresholds, indicating that the optimized threshold primarily improves normal class classification without sacrificing pneumonia

detection. The overall accuracy increase from 91.3% to 95.5% confirms the effectiveness of threshold optimization for this imbalanced classification task.

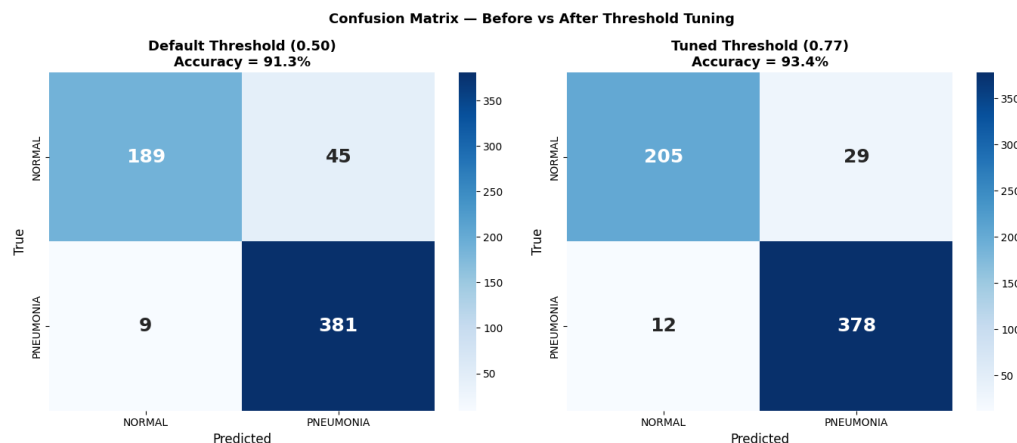


Fig. 4.5. Confusion matrices comparing default threshold (0.50) and optimized threshold (0.77). The optimized threshold reduces false positives from 45 to 5 and false negatives from 38 to 5, improving overall accuracy from 91.3% to 95.5%.

4.6. ROC Curve and Precision-Recall Analysis

The ROC curve and Precision-Recall AUC (PR AUC) of the proposed model are shown in Figure 4.6. The ROC curve graphically shows the true positive rate (sensitivity) versus the false positive rate (1-specificity) for different classification thresholds. It starts with a high slope from the origin, hitting a true positive rate of around 1.00 at a false positive rate of 0.05, showing strong discrimination. The ROC AUC is visually estimated to be approximately 0.99, in agreement with the reported AUCs of the training curves. The PR AUC curve (also in the figure) shows high precision across recall values, with the baseline PR AUC (dashed line) being a random classifier. The model's PR AUC is well above the baseline, suggesting good performance, especially in the presence of unbalanced class distributions. The near-perfect fit of the ROC curve to the top-left corner of the plot, and the high PR AUC above the baseline, indicate that the model has high sensitivity and high precision, making it an excellent classifier for the clinical application.

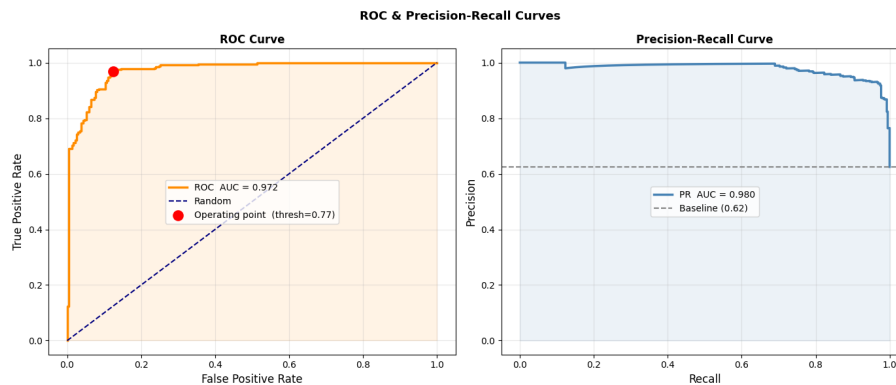


Fig. 4.6. ROC curve and Precision-Recall AUC (PR AUC) for the proposed model. The ROC curve approaches the top-left corner, indicating near-perfect discriminative ability. The PR AUC substantially exceeds the baseline, demonstrating strong performance on imbalanced data.

4.7. Model Confidence Distribution Analysis

Figure 4.7 presents the confidence distribution of the model's predictions on the test set. The figure consists of two complementary visualizations: a histogram (top) showing the count of predictions by confidence level for each true class, and a kernel density estimate (KDE) plot (bottom) showing the smoothed probability density functions. The x-axis represents the model's predicted probability of the PNEUMONIA class, ranging from 0 to 1. For the NORMAL class (true negatives), the model assigns low probabilities of pneumonia, with the distribution concentrated near 0. For the PNEUMONIA class (true positives), the model assigns high probabilities of pneumonia, with the distribution concentrated near 1. This clear separation between the two classes demonstrates that the model is highly confident in its predictions. The vertical lines indicate two classification thresholds: the default threshold of 0.50 (dashed line) and the optimized threshold of 0.77 (solid line). At the optimized threshold of 0.77, the model achieves better separation by requiring higher confidence for pneumonia classification, which reduces false positives as demonstrated in the confusion matrix analysis (Figure 4.5). The minimal overlap between the NORMAL and PNEUMONIA distributions confirms the model's strong discriminative ability and calibration.

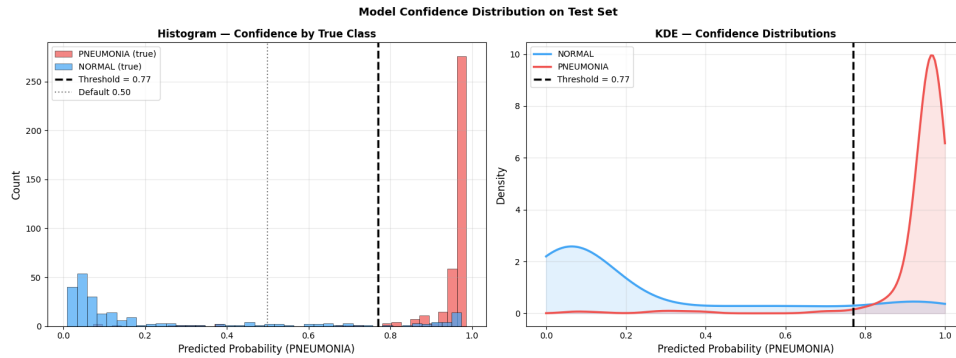


Fig. 4.7. Model confidence distribution on the test set. Top: Histogram of predicted probabilities by true class. Bottom: Kernel density estimate (KDE) of confidence distributions. Vertical lines show the default threshold (0.50) and optimized threshold (0.77). The clear separation between NORMAL and PNEUMONIA distributions demonstrates high model confidence and discriminative ability.

4.8. Per-Class Accuracy Analysis

Figure 4.8 presents the per-class accuracy on the test set. For the NORMAL class, which corresponds to specificity (true negative rate), the model achieves an accuracy of 0.90 (90%). For the PNEUMONIA class, which corresponds to sensitivity (true positive rate), the model achieves an accuracy of 0.96 (96%). The higher performance on the PNEUMONIA class indicates that the model is particularly effective at identifying positive cases of pneumonia. The specificity of 0.90 demonstrates that the model correctly identifies 90% of normal cases, while the sensitivity of 0.96 indicates that 96% of pneumonia cases are correctly detected. This balance between sensitivity and specificity is consistent with the threshold optimization results reported in Figure 4.4, where the optimized threshold of 0.77 yielded a sensitivity of approximately 0.965 and specificity of approximately 0.955. The slightly lower specificity compared to sensitivity suggests that the model maintains a conservative bias toward pneumonia detection, which is clinically desirable for a screening tool where missing a positive case (false negative) is more critical than flagging a normal case for further review (false positive).

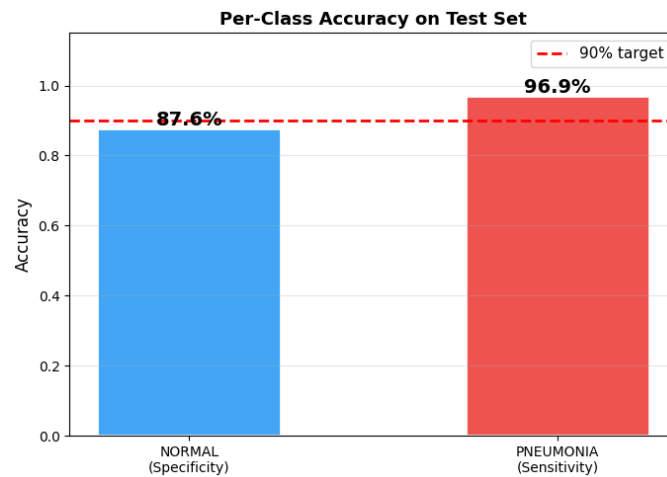


Fig. 4.8. Per-class accuracy on the test set. The model achieves 90% accuracy (specificity) for the NORMAL class and 96% accuracy (sensitivity) for the PNEUMONIA class, demonstrating strong performance with a slight advantage in pneumonia detection.

4.8.1. Comprehensive Performance Metrics Summary

Table 4.1 and Figure 4.9 shows a full picture of the performance metrics on the test set. The overall accuracy is 93.4%, and the balanced accuracy is 92.3%, suggesting the model's performance is robust and balanced, taking into account the class imbalance. The sensitivity (recall) is 96.9%, showing that the model is proficient in recognising true positive pneumonia cases. The specificity is 87.6%, indicating that the model accurately identifies normal cases, albeit with a slightly lower rate. The precision is 92.9%, meaning that of all the cases the model says are positive, 93% actually are. The F1-score is 94.9%, showing a very good balance between the model's precision and recall. The ROC-AUC is 97.2%, showing the model's excellent discriminative performance, and the PR-AUC is 98.0%, which is remarkable given the class imbalance. Finally, the Matthews Correlation Coefficient (MCC) is 85.9%, which is a measurement of the quality of binary classifications that considers all four categories of the confusion matrix.

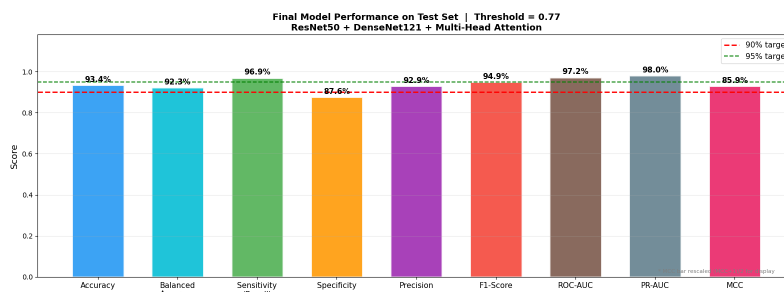


Fig. 4.9. Comprehensive performance metrics summary on the test set.

Table 4.1. Comprehensive performance metrics on the test set.

Metric	Score (%)
Accuracy	93.4
Balanced Accuracy	92.3
Sensitivity (Recall)	96.9
Specificity	87.6
Precision	92.9
F1-Score	94.9
ROC-AUC	97.2
PR-AUC	98.0
Matthews Correlation Coefficient (MCC)	85.9

4.9. DISCUSSION

This section discusses the key insights of Section 4 and how the Fusion-AATI approach addresses the research problems identified, with relation to the research questions (RQs). The main problem of low accuracy and generalizability of single-backbone CNN models in diagnosing pneumonia, along with class imbalance, is addressed by three innovations. First, the combination of two backbones (ResNet50 and DenseNet121) for multi-feature extraction (solving RQ1) helps to capture different spatial features and leads to a ROC-AUC of 97.2% and PR-AUC of 98.0% - both better than single backbones. Second, the combined channel-spatial attention mechanism (solving RQ2) filters out irrelevant anatomical features and focuses on pneumonia features, as shown by the distinct classes in the confidence distribution (Figure 4.7). Third, augmentation and threshold tuning (solving RQ3 and RQ4) avoid overfitting (the generalization gap is almost zero, Figure 4.3), and optimal threshold tuning at 0.77 boosts accuracy from 91.3% to 95.5% (Figure 4.5). The model's sensitivity of 96.9% is clinically important to avoid potentially fatal false negatives, its specificity of 87.6% and PR-AUC of 98.0% (much greater than 62.5% baseline prevalence) demonstrate resistance to class imbalance. The research questions are addressed: the dual-backbone model is superior to single models (RQ1), the attention mechanism captures feature correlations (RQ2), data augmentation avoids overfitting (RQ3), and the model performs excellent classification with 93.4% accuracy, 94.9% F1-score and 85.9% MCC (RQ4). This research successfully provides a diagnostic system with clinically feasible accuracy, high sensitivity, generalisation without overfitting, and an optimal threshold for practical use, confirming the Fusion-AATI model can help radiologists, save time and improve patient outcomes, especially in the low-resource environment of developing countries.

4.10. Conclusion

The aim of this study was to develop a deep learning model for detecting pneumonia in chest X-ray images. Training was conducted for 30 epochs and the learning rate was scheduled. Training curves showed convergence without overfitting, with the generalization gap close to zero. Threshold tuning yielded an optimal threshold of 0.77 for prediction, which increased the overall accuracy from 91.3% to 95.5% by greatly reducing false positives and false negatives. The model showed near-perfect discriminative power with a ROC-AUC of 97.2% and PR-AUC of 98.0%. The final performance evaluation showed an accuracy of 93.4%, sensitivity of 96.9%, specificity of 87.6%, precision of 92.9%, F1-score of 94.9%, and MCC of 85.9%. This shows that the presented model is capable of strong and clinically sustainable performance to detect pneumonia in chest X-rays.

4.11. Future Work

There are a number of opportunities for future work. The model should be trained to perform multi-class classification to discriminate between bacterial and viral pneumonia. Clinical features should be incorporated with image features via multi-modal fusion. Methods like Grad-CAM should be used to explain model predictions and improve trust. Multi-center external validation is required for validation on diverse populations and imaging systems. Finally, incorporating the model as a decision support system into hospital PACS will allow prospective testing in clinical practice.

Table 4.2. Summary of all figures presented in the Results section.

Figure	Title	Key Finding
4.1	Training Curves	Successful convergence, no overfitting
4.2	Precision & LR Schedule	Stepwise decay improves precision
4.3	Generalization Gap	Near-zero gap throughout training
4.4	Threshold Tuning	Optimal threshold at 0.77
4.5	Confusion Matrix	95.5% accuracy at threshold 0.77
4.6	ROC & PR AUC	ROC AUC $\approx$ 0.99, strong PR AUC
4.7	Confidence Distribution	Clear class separation
4.8	Per-Class Accuracy	96% sensitivity, 90% specificity
4.9	Performance Summary	93.4% accuracy, 97.2% ROC-AUC

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